Math 67 – HW 3 Solutions

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Chapter 5

C2. Given vector space $\mathbb{C}^3$ and set $\{v_1, v_2, v_3\} \in \mathbb{C}^3$, where vectors

$$v_1 = (i, 0, 0), \quad v_2 = (i, 1, 0), \quad v_3 = (i, i, -1) \quad \in \mathbb{C}^3$$

(a) $\text{span}\{v_1, v_2, v_3\} = \mathbb{C}^3$.

Proof. Consider $(x, y, z) \in \mathbb{C}^3$ such that

$$(x, y, z) = a_1v_1 + a_2v_2 + a_3v_3 = (a_1i, 0, 0) + (a_2i, a_2, 0) + (a_3i, a_3i, -a_3)$$

for some $a_1, a_2, a_3 \in \mathbb{C}$. Component-wise,

$$z = -a_3, \quad y = a_2 + a_3i, \quad x = a_1i + a_2i - a_3i.$$

Then

$$a_3 = -z, \quad a_2 = y + zi, \quad a_1 = -(z + y) - (x + z)i. \tag{1}$$

$\mathbb{C}^3 \subseteq \text{span}\{v_1, v_2, v_3\}$: Any $(x, y, z) \in \mathbb{C}^3$ is in the span of $\{v_1, v_2, v_3\}$ because we can express it as the linear combination $a_1v_1 + a_2v_2 + a_3v_3$ where a_1, a_2, a_3 are given by (1). $\text{span}\{v_1, v_2, v_3\} \subseteq \mathbb{C}^3$: each $v_j \in \mathbb{C}3$ and C^3 is a vector space. □

(b) v_1, v_2, and v_3 form a basis for $\mathbb{C}^3$.

Proof. Part(a) shows that v_1, v_2, and v_3 span $\mathbb{C}^3$, so we only need to show that the vectors are linearly independent: for $(x, y, z) = (0, 0, 0) = 0 \in \mathbb{C}^3$, equation (1) implies that we must have coefficients $a_1 = a_2 = a_3 = 0$. □

C3.

(a) For $V = \{ (x_1, x_2, x_3, x_4) \in \mathbb{F}4 \mid x_4 = 0 \}$, $\dim(V) = 3$.
Only 3 linearly independent vectors. 0 is always linearly dependent.

(b) $V = \{ (x_1, x_2, x_3, x_4) \in \mathbb{F}4 \mid x_4 = x_1 + x_2 \}$. $\dim(V) = 3$. All the information in this space can be provided by x_1, x_2, x_3.

(c) $V = \{ (x_1, x_2, x_3, x_4) \in \mathbb{F}4 \mid x_4 = x_1 + x_2, x_3 = x_1 - x_2 \}$. $\dim(V) = 2$.

(d) $V = \{ (x_1, x_2, x_3, x_4) \in \mathbb{F}4 \mid x_4 = x_1 + x_2, x_3 = x_1 - x_2, x_3 + x_4 = 2x_1 \}$. $\dim(V) = 2$
x_3 and x_4 can be obtained with x_1 and x_2.

(e) $V = \{ (x_1, x_2, x_3, x_4) \in \mathbb{F}4 \mid x_1 = x_2 = x_3 = x_4 \}$. $\dim(V) = 1$

C4. We want values for λ that make the following sets linearly dependent.

(a) Let $\{v_1, v_2, v_3\} = \{ (\lambda, -1, -1), (-1, \lambda, -1), (-1, -1, \lambda) \} \subset \mathbb{R}^3$.
A quick look suggests to set $\lambda = -1$. Then this set is linearly dependent. Stated formally,

$$1v_1 - 1v_2 + 0v_3 = (0, 0, 0) = 0 \in \mathbb{C}^3,$$

i.e. there exists a non-trivial, namely $(a_1, a_2, a_3) = (1, -1, 0)$ solution to the homogeneous (equaling zero) equation $a_1v_1 + a_2v_2 + a_3v_3 = 0$.

(b) Let $\{v_1, v_2, v_3\} = \{ \sin^2 x, \cos 2x, \lambda \} \in \mathcal{C}(\mathbb{R})$.
Recall the trigonometric identity (double-angle formula) $2\sin^2\theta + \cos 2\theta = 1$ for any θ. Then the constant function $\lambda = 1 \in \mathcal{C}(\mathbb{R})$ makes $\{v_1, v_2, v_3\}$ linearly dependent.

P1. Let V be a vector space over $\mathbb{F}$ and suppose the set of vectors $\{v_1, v_2, ..., v_n\}$ spans V, where each $v_i \in V$. Then

$$\{v_1 - v_2, v_2 - v_3, v_3 - v_4, ..., v_{n-2} - v_{n-1}, v_{n-1} - v_n, v_n\} \tag{$\star$}$$

also spans V.

Proof. For any $x \in \text{span}\{v_1, v_2, \ldots, v_n\}$,

$$\begin{aligned} x &= a_1v_1 + a_2v_2 + \cdots + a_nv_n \\ &= b_1v_1 + (b_2 - b_1)v_2 + \cdots + (b_n - b_{n-1})v_n \\ &= b_1(v_1 - v_2) + \cdots + b_{n-1}(v_{n-1} - v_n) + b_nv_n \end{aligned}$$

where $b_1 = a_1, b_2 = a_2 + b_1, b_3 = a_3 + b_2, \ldots, b_n = a_n + b_{n-1}$. The existence of scalars $a_j \in \mathbb{F}$ implies the existence of scalars $b_j \in \mathbb{F}$ and vice versa. Any $x \in V$ can be expressed as a linear combination of vectors from either set. Thus, if $\{v_1, v_2, \ldots, v_n\}$ spans V then ($\star$) must also span V. □

P2. Let V be a vector space over $\mathbb{F}$ and suppose that $\{v_1, v_2, ..., v_n\}$ is a linearly independent set of vectors in V. Suppose we have a $w \in V$ such that $\{v_1 + w, v_2 + w, ..., v_n + w\}$ is linearly *dependent*. Then w is in the span of $v_1, v_2, ..., v_n$.

Proof. Since the set $\{v_1 + w, v_2 + w, ..., v_n + w\}$ is linearly dependent, there exist scalars $a_0, a_1, \ldots, a_n$ (at least one of which is nonzero) such that

$$a_1(v_1 + w) + a_2(v_2 + w) + \cdots + a_n(v_n + w) = 0.$$

Then

$$w = -\frac{a_1v_1 + a_2v_2 + \cdots + a_nv_n}{\sum_j a_j}$$

is a linear combination of vectors v_j, and thus $w \in \text{span}\{v_1, v_2, \ldots, v_n\}$. □

P4. Let V be a vector space over $\mathbb{F}$ and let U be a subspace of V such that $\dim(U) = \dim(V)$. Then $U = V$.

Proof. Let n be the dimension of the spaces U and V and let the set $\{u_j\}_{j=1}^n = \{u_1, u_2, \ldots, u_n\}$ be a basis for U. Since U is a subspace of V, it follows that $U \subseteq V$; in particular, each basis element $u_j \in V$, so the set $\{u_j\} \subseteq V$. Furthermore, $\{u_j\}$ is a linearly independent set (by definition of a basis), and since it has length n, it follows from Theorem 5.4.4 (part3) that it must be a basis for V. Then $V = \text{span}\{u_j\} = U$. $\square$

P5. $\mathbb{F}_m[z]$ is the vector space of all polynomials of degree m or less, with coefficients in $\mathbb{F}$. Suppose that $p_0, p_1, ..., p_m \in \mathbb{F}_m[z]$ all satisfy $p_j(2) = 0$. Then the set of vectors $\{p_0, p_1, ..., p_m\}$ is linearly dependent.

Proof. (by contradiction). Suppose that $\{p_0, p_1, ..., p_m\} \subset \mathbb{F}_m[z]$ is a linearly independent set. The dimension of $\mathbb{F}_m[z]$ is $m+1$ (see example 5.4.3), and since there are $m+1$ linearly independent vectors p_j, we must have that $\text{span}\{p_0, p_1, ..., p_m\} = \mathbb{F}_m[z]$. Now consider the function $f(x) = 1$ which is a polynomial of degree 0, so $f \in \mathbb{F}_m[z]$. Then f must be in the span of $p_0, p_1, ..., p_m$, i.e. so there are some coefficients $a_j \in \mathbb{F}$ such that

$$1 = \alpha_0 p_0(z) + \alpha_1 p_1(z) + \cdots + \alpha_m p_m(z)$$

for all z. But since $p_j(2) = 0$ for every j, when $z = 2$ we have $1 = 0$, which is absurd.

Our supposition that $\{p_0, p_1, ..., p_m\}$ is a linearly independent set must be false. Hence, $\{p_0, p_1, ..., p_m\}$ is linearly dependent. $\square$

P6. Let U and V be five-dimensional subspaces of $\mathbb{R}^9$. Then $U \cap V \neq \{0\}$.

Proof. Let $\{\, u_1, u_2, u_3, u_4, u_5 \,\} \subset U$ be a basis for U and let $\{\, v_1, v_2, v_3, v_4, v_5 \,\} \in V$ be a basis for V (we know these bases exist because U and V are five dimensional). The set $\{u_1, u_2, u_3, u_4, u_5, v_1, v_2, v_3, v_4, v_5\} \subset \mathbb{R}^9$ must be linearly dependent because there are ten of them and $\dim(\mathbb{R}^9) = 9$. (It takes only 9 linearly independent vectors to span $\mathbb{R}^9$.) Then one of the basis vectors v_j of V must be in U. Thus we have that $v_j \in U \cap V$. $\square$